Fiscal Year:	FY 2025	**Task Last Updated:**	FY 01/10/2025
PI Name:	Selva, Daniel Ph.D.		
Project Title:	HCAAM VNSCOR: Virtual Assistant for Spacecraft Anomaly Treatment During Long Duration Exploration Missions		
Division Name:	Human Research		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:			
Joint Agency Name:		**TechPort:**	Yes
Human Research Program Elements:	(1) **HFBP**:Human Factors & Behavioral Performance (IRP Rev H)		
Human Research Program Risks:	(1) **HSIA**:Risk of Adverse Outcomes Due to Inadequate Human Systems Integration Architecture		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Zip Code:	77843-0001	**Congressional District:**	17
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2017-2018 HERO 80JSC017N0001-BPBA Topics in Biological, Physiological, and Behavioral Adaptations to Spaceflight. Appendix C
Start Date:	03/06/2019	**End Date:**	08/31/2025
No. of Post Docs:	0	**No. of PhD Degrees:**	1
No. of PhD Candidates:	2	**No. of Master' Degrees:**	0
No. of Master's Candidates:	1	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	1	**Monitoring Center:**	NASA JSC
Contact Monitor:	Whitmire, Alexandra	**Contact Phone:**	
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Flight Program:			
Flight Assignment:	NOTE: End date changed to 08/31/2025 per A. Beitman/JSC (Ed., 9/19/24) NOTE: End date changed to 02/28/2025 per A. Beitman/JSC (Ed., 1/25/23)		
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Diaz Artiles, Ana Ph.D. (Texas A&M Engineering Experiment Station) Dunbar, Bonnie Ph.D. (Texas A&M Engineering Experiment Station) Wong, Raymond Ka Wai Ph.D. (Texas A & M, College Station)		
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Task Description:	This task is part of the Human Capabilities Assessments for Autonomous Missions (HCAAM) Virtual NASA Specialized Center of Research (VNSCOR). The research objective of this proposal is to investigate the impact of using Virtual Assistants (VA) to support crewmembers in the context of anomaly treatment during Long Duration Exploration Missions (LDEM), when ground support will be limited. A VA will be developed building upon the software architecture from existing VAs developed by the Principal Investigator (PI) for similar purposes. The VA will provide support for various aspects of anomaly treatment, including detecting and diagnosing the anomaly, as well as recommending a course of action. It will also have the ability to take initiative in the dialog with the user (mixed-initiative mode), and the ability to provide explanations for its actions. The impact of the VA on performance, cognitive workload, situational awareness, and trust, will be assessed through a set of three experiments with human subjects in a laboratory environment. The first experiment will establish the baseline impact (master-slave, no explanations), and subsequent experiments will study the effect of switching to the mixed-initiative mode and adding explanations. The system will also be deployed and tested in the Human Exploration Research Analog (HERA) analog environment.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	This project will provide standards and guidelines that will help NASA design similar virtual assistants to support astronauts during future long duration exploration missions. Such standards and guidelines will concern both the functionality and the user interface of the virtual assistant.
Task Progress:	In Year 6 of this project, we have worked primarily on Specific Aim 3, supporting the conduct of the HERA C7 campaign. The goal of Campaign 7 was to test a more advanced version of the assistant with more complex anomaly scenarios. In addition, we have worked on publications regarding Specific Aims 1 and 2. Finally, we started working on some guidelines for the future development of similar VAs. Specific Aim 3: HERA C7 campaign Concerning Specific Aim 3, we prepared for and supported the HERA C7 campaign, which was completed in December 2024. Data analysis is in progress as of January 10, 2025. In preparation for the campaign, we conducted two major tasks: We added several new “complex” anomaly scenarios, consisting of two and in some cases 3 anomalies. These anomalies were cascading in most cases, with the second one being phased in a few seconds or minutes after the first one. In a few cases, the individual anomalies were simultaneous. These scenarios were meant to simulate both independent and common-cause failures, as well as cases where the first failure causes the second one. The new scenarios were designed by HERA personnel with input from us and were largely combinations of existing (C6) scenarios. The design of these anomaly scenarios was subject to operational constraints on duration, number of operators, etc. Developing these scenarios necessitated incorporating them into Daphne’s knowledge graph and making changes to the Daphne diagnosis skill to allow the calculation of likelihood scores for these complex scenarios. We also used the “enhanced” version of Daphne in C7. Compared to the “baseline” version used in C6, the enhanced version had the following added functionalities: 1) provide explanations for the likelihood scores of each anomaly; 2) a more advanced question-answering system leveraging large language models (GPT4). With data from the first 3 missions (N=12), the major findings from the HERA C7 campaign are relatively similar to those of the C6 campaign: –No significant differences in any of the main measures –Time to diagnosis (min): 9.29±1.48 (without VA) vs 8.51±1.73 (with VA) –Situational Awareness Rating Technique (SART) 25.33±2.10 (without VA) vs 24.69±1.65 (with VA) –NASA Task Load Index (TLX) 30.13±3.21 (without VA) vs 29.02±2.99 (with VA) Insights from the exit interviews are largely consistent with those of C6. By and large, subjects thought the scenarios became easy and repetitive towards the end of the mission, due to a lack of variety. Indeed, many scenarios designed by HERA were different combinations of the same few C6 scenarios (CDRA failure, cabin fan failure, MOXIE failure). All subjects established trust in Daphne very quickly and cited the tool’s accuracy as the primary factor for this. Some subjects also identified Daphne’s ability to provide explanations for its diagnoses as a positive factor that contributed to the establishment of trust. Almost none of them found the question-answering capabilities essential because the task was clear, they were trying to finish the task as quickly as possible, and they didn’t feel like they needed to ask any questions. Some subjects did mention that question-answering was more useful in cases where Daphne recommended more than one diagnosis with the same confidence level, and thought it would be even more useful if the anomaly was completely new and unknown. Overall, crewmembers felt that in more complex scenarios, a more iterative approach to diagnosis with Daphne asking questions to the crewmembers to try to gather more evidence and refine its diagnosis would be useful. Note that detailed data regarding the interactions between the crewmembers and Daphne has not been analyzed yet. Guidelines We have developed a list of 20+ potential guidelines. Each guideline consists of a short title, a recommendation, and some justification/evidence. An example guideline is provided below. Short title: On-demand adaptive explanations Recommendation: AI agents should give succinct or on-demand explanations for users who already know and trust the tool and longer explanations for more novice users who are expected to not trust the tool yet. Justification: More explanations for novices can accelerate training and develop trust, but long explanations can be frustrating and lead to a higher workload for advanced users (Bansal et al., 2021). In C7 exit interviews, several crewmembers mentioned that they don’t need a symptom comparison table because they already trust the model and they want to go as fast as possible. The complete list of current guidelines is provided below. This list is still in progress and the guidelines are being more fully developed. –Ranked anomaly scenarios: Provide a list of plausible anomaly causes ranked by likelihood –Symptom comparison

	table: Show a table comparing the signature of current vs potential anomalies –Diagnosis confidence: Agent should convey the degree of uncertainty in its diagnosis –Anomaly signature changes: Alert the user when the signature of the anomaly has changed –Automatic TM display: Automatically pull up time series for relevant parameters when an anomaly is detected –Potential misdiagnosis: Alert the user when asking to diagnose symptoms that were likely created by user actions –Agent information space: AI agent should have access to 1) the telemetry feed; 2) a database of the most recent N days of telemetry; 3) a knowledge base of subsystems, anomalies, procedures; 4) a historical anomaly database; 5) information about its own internal workings –Voice interface: Agent should have a voice interface to help the users, particularly with following procedures –Sufficient training: For proper agent trust, users should receive both hands-on training and theoretical training on 1) the tool and how it works internally; 2) the habitat systems –Latency of the QA system: The QA system shall be fast enough to provide immediate answers –Flexibility of the QA system: Use an LLM to increase flexibility and scalability. However, it should be a local LLM that can be downloaded and fine-tuned on a relevant dataset –Top-down Workflow: Users find a top-down workflow (detection, diagnosis, and resolution) intuitive, with the chatbot on the right –Evidence from other agents: Agent should incorporate additional evidence provided by the user and provide an updated diagnosis –Bayesian diagnosis: Agent should be able to update initial beliefs based on all evidence provided –Propose actions to reduce diagnostic uncertainty: Agent should recommend actions to gather additional evidence to refine the diagnosis in case of uncertainty –Agent learning: Agent should be able to learn from its own experience –Explanation types: The agent should provide explanations based on both expert knowledge and historical anomaly occurrence –Outlier detection: The agent should flag outlier signatures even if no thresholds are exceeded –Multiple anomalies: The agent should be able to deal with multiple simultaneous anomalies –Anomaly priority: In case of multiple anomalies, the agent should help the user prioritize the most important ones to attend to first –Full-wave diagnosis: The agent should diagnose anomalies based on the current and past values of the parameters in the telemetry feed (as opposed to the thresholds exceeded)
Bibliography Type:	Description: (Last Updated: 12/17/2025)
Abstracts for Journals and Proceedings	Selva D, Dutta P, Josan PK, Dunbar BJ, Wong RKW, Diaz-Artiles A. "Virtual assistant for anomaly resolution in long duration exploration missions: Final results and lessons learned." 2025 NASA Human Research Program Investigators' Workshop, Galveston, Texas, January 28-31, 2025. Abstracts. 2025 NASA Human Research Program Investigators' Workshop, Galveston, Texas, January 28-31, 2025. , Jan-2025
Articles in Other Journals or Periodicals	Dutta P, Josan PK, Wong RKW, Dunbar BJ, Diaz-Artiles A, Selva D. "Effects of explanations and accuracy on human performance and trust in AI-assisted anomaly diagnosis tasks." Journal of Cognitive Engineering and Decision Making. Conditionally accepted November 14, 2024. , Nov-2024
Articles in Peer-reviewed Journals	Josan PK, Dutta P, Abbott R, Martin AV, Dunbar BJ, Wong RKW, Selva D, Diaz-Artiles A. "Virtual assistant for spacecraft anomaly resolution: effects on human performance metrics." J Aerosp Inform Syst. 2025 Apr. 22(4):264-74. https://doi.org/10.2514/1.I011449 , Apr-2025